

TEACHER GUIDE

Folding Camping Shovel · teacher guide

Teaching guidance, answers and evidence criteria for the 70-page student edition

What this book covers

The student workbook covers the five guided project modules, all 15 named theory sections, all 150 original multiple-choice questions and 450 options, all 15 written evidence tasks, the twelve-part project folio and the six supplied original plan sheets. The separate answer guide has an appropriate length and is not part of the student's 70-page total.

Use the book with the current project sources

The printed explanations support reasoning and evidence. The current teacher-issued brief, drawings, SOPs, WMS, demonstrations and permissions control practical production. Confirm the local material information, settings, methods, acceptance criteria and authorised changes before work. A completed response or checklist never grants machine permission.

Judge evidence and reasoning

Multiple-choice answers are source-defined. Open work can have several valid answers: assess traceability, correct concepts, reasoning and the student's actual contribution. A hypothetical example is labelled as such. Planned observations and pending checks must not be presented as completed practical evidence. Reward clear limits rather than confident unsupported claims.

Print and work on paper

The controlled student PDF is 70 A4 pages, including cover and contents, equivalent to 35 sheets printed double-sided on the long edge. The six original drawing reproductions use landscape pages within the book; preserve orientation and use actual-size printing. Never scale a resized drawing image to obtain a missing dimension. Word text, questions, tables and response fields remain editable.

Use the answer references

Answers are keyed to module, question number and student page. Exact written tasks retain their source wording. Original folio prompts have linked response, caption and source/checkpoint criteria. A shared piece of evidence may support several prompts only if it addresses all of their requirements explicitly. Review the actual work, not only the number of completed fields.

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Authority, scope and current confirmation

Keep the historical originals and current teaching responsibilities distinct.

The six original drawings

The WWHS plan package is dated 4/11/14. Preserve all drawing information. The assembly sheet, handle end components, head flat pattern, folded head, pivot components and pivot-angles assessment sheet form a coordinated source set. Compare with the current issued version before use. The original assessment sheet's 60-minute and 50-mark panel is historical source content; it does not set this workbook's current assessment conditions.

Assessment and timetable

The public Task 2 notice is a draft and contains older assessment/outcome metadata. The current v2.1 teaching program uses a later syllabus basis and retains local confirmation fields. The student book therefore does not issue a weighting, calendar date, component marks or formal outcome mapping. Use the current teacher-issued notice. Five modules describe the learning route; they are not a promise of the source cards' ten weeks or the program's sixteen weeks.

A bounded course companion

The book follows the fifteen named sections in the guided Folding Camping Shovel site. The shared syllabus-learning hub is a broader cross-project extension; this workbook does not claim to reproduce every hub activity or the full 200-hour Metal course. Teachers can use that hub separately where their approved program requires it.

Original arithmetic practice

The metric drilling relationship $n = (1000 \times v_c) \div (\pi \times D)$ is supported by Sandvik Coromant's primary formula reference. The printed 20 m/min with 10 mm example and 15 m/min with 12 mm practice task are invented arithmetic inputs only. They are not material recommendations, drill selection or authorised machine settings. The actual project calculation uses teacher-supplied data and teacher-confirmed settings.

Illustrations and real evidence

New photographs are illustrative studies generated for teaching, not photographs of a student's completed work. They support observation and discussion, not construction approval. Original plans retain their source authority. Students should supply their own authorised production evidence and avoid publishing personal information or photographs of other students without permission.

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Source route and module crosswalk

The source section titles remain visible in the workbook.

Why the editorial labels differ

Several website navigation labels no longer describe the named sections they contain. The book retains Module 1–5 and all exact section titles, while its running labels describe the actual learning. The crosswalk below lets a teacher find the matching website module without importing the misleading label into the lesson explanation.

Module / workbook label	Source navigation label
1. Drawings, planning and impacts	Brief, modifications and job planning
2. Shovel-head production	Risk, WMS and time/action planning
3. Lathe, fits and threads	Alloys and shovel-head production
4. Pivot, milling and RPM	Fits, threads and machining
5. Assembly and evaluation	Pivot, RPM, assembly and evaluation

Course and original sources

Course: <https://stevencowell.github.io/Yr-10-Metal/folding-camping-shovel/> . Original drawing package: project-resource.html within that site. Source snapshot: Git revision d2ccaf84ff7cf8d46a4ee6421fa39d676c88f7fa, frozen 25 September 2026. The course content, lesson runtime, five presentations and six original PDFs were checked against their live bytes.

Optional supporting media

The matching five PowerPoint presentations have eight slides each. The source-linked video ‘Malleability and Ductility-Physical Properties’ by MooMooMath and Science can support section 11.2: <https://www.youtube.com/watch?v=CIBXoYaM7Fw> . Ask students to distinguish the two terms, then apply them only to confirmed material and process evidence. Essential print explanations do not depend on watching the video.

Reference and image credits

Primary drilling formula reference: Sandvik Coromant, Formulas and definitions for drilling, metric. Original drawings: WWHS, 4/11/14, recovered from the teacher-supplied project source package. Original source course text and questions are used within the authorised course workbook. New illustrations were generated through the available image-generation tool; the tool did not expose a model version. No photograph is claimed as an actual student product.

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Module 1 • answers 1–10

Student pages 12–13 / Drawings, planning and impacts

- 1. A. Show how major parts connect** — Assembly drawings explain how major parts relate, while detail drawings provide information about individual components.
- 2. B. To understand complete component relationships** — Comparing multiple drawing views helps reveal relationships that cannot be understood from a single image alone.
- 3. B. The written dimension** — Written dimensions and notes are authoritative, while appearance should never override documented drawing information.
- 4. B. Internal or hidden relationships** — Sectional views expose hidden features and relationships that are not visible from external views alone.
- 5. A. Components form an interconnected system** — A change to one component can influence another because the product operates as an interconnected system.
- 6. C. Information controlling position, movement or fit** — Students should identify the information that governs position, movement and fit before practical work begins.
- 7. A. Component identity and drawing revision** — Checking component identity and revision helps ensure production follows the current approved drawing version.
- 8. B. Record the issue and refer it to the teacher** — Recording uncertainties and referring them creates an evidence trail and avoids unsupported assumptions.
- 9. C. Written dimensions and notes are the authoritative source** — Drawing appearance can mislead, so written dimensions and notes remain the authoritative production reference.
- 10. A. Reliable production planning based on verified information** — Tracing components across related views builds a reliable evidence trail from drawing interpretation to production planning.

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Module 1 • answers 11–20

Student pages 14–15 / Drawings, planning and impacts

11. A. Coordinate the authorised construction processes — A whole-product WMS records the authorised sequence and links processes rather than inventing new methods or replacing teaching.

12. B. Dependencies between stages — Showing dependencies explains why one stage must be completed or checked before later work can continue safely and correctly.

13. A. To prevent unverified work progressing — Quality hold points stop work continuing until important results have been verified before later stages depend on them.

14. C. The WMS coordinates the overall project while the SOP covers the lathe process — The WMS explains where lathe work fits into the project, while the SOP governs approved lathe operation.

15. A. Task, hazards, controls, permissions and quality hold points — Each WMS stage records the task together with approved safety, permission and quality information needed for that step.

16. B. Wait until the result is verified before dependent work continues — Hold points exist so later work does not rely on results that have not yet been confirmed.

17. C. Stop, report the change and update the WMS only with teacher direction — Changes require students to stop, report them and revise the WMS only under teacher direction.

18. A. Neither replaces teacher demonstration, supervision or authorisation — Written planning supports but never replaces teacher demonstration, supervision and authorisation during practical work.

19. B. To show which stages rely on earlier completed and inspected work — Visible dependencies help students recognise when later tasks rely on earlier verified results before proceeding.

20. C. The current teacher-issued drawings, SOPs, WMS template and demonstrations — The source boundary states that current authorised documents and demonstrations determine unresolved process and quality details.

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Module 1 • answers 21–30

Student pages 16–17 / Drawings, planning and impacts

- 21. A. Because some production decisions are difficult to reverse later** — Considering impacts early helps identify consequences before production choices become difficult or impractical to change.
- 22. C. How durable and repairable is the product expected to be?** — Durability and repairability are evidence-based impact criteria that help evaluate long-term product performance and use.
- 23. A. The confirmed folding-camping-shovel context and reliable evidence** — Societal claims should reflect the confirmed project context and verified evidence rather than unsupported assumptions.
- 24. B. Consider resources, waste, maintenance and end-of-life outcomes** — Environmental criteria examine the product's lifecycle, including resource use, waste, maintenance and end-of-life considerations.
- 25. C. Such claims require verified supporting information** — Broad environmental claims need reliable evidence before they can be presented as justified conclusions.
- 26. A. Improving durability may require more resources or processing** — Trade-offs recognise that improving one aspect may create costs or disadvantages in another area.
- 27. B. Explain what evidence is still needed before reaching conclusions** — Strong reports identify remaining evidence needs instead of presenting uncertain conclusions as established facts.
- 28. C. Responsible long-term use should be considered using reliable evidence** — Responsible long-term use should be evaluated from verified evidence within the confirmed project context.
- 29. A. The teacher-issued brief, drawings, approved research sources and demonstrations** — Unresolved impact details must be confirmed using the authorised project sources rather than guesses or assumptions.
- 30. B. Durability and resource use** — The theory explains that improving durability may increase resource use or manufacturing effort, requiring balanced judgement.

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Module 2 · answers 1–10

Student pages 22–23 / Shovel-head production

- 1. A. The correct component and current drawing revision** — Correct planning begins by confirming the approved drawing information before any production decisions are made.
- 2. B. To avoid relying on assumptions about typical shovel construction** — The demonstration supports the authorised interpretation instead of assumptions about how similar products are usually manufactured.
- 3. C. To provide consistent reference points for locating and checking features** — Datums create reliable references so features can be positioned and checked consistently throughout production.
- 4. A. The relationship between faces, edges and the folding assembly** — Orientation marks maintain the intended relationship between the component and the rest of the assembly.
- 5. B. To recognise that some features depend on earlier verified work** — Understanding dependencies helps students recognise which features rely on earlier verified work without changing the approved sequence.
- 6. C. Before irreversible stages and before later work depends on the result** — Inspection before critical stages helps detect problems before later work depends on inaccurate results.
- 7. A. The component's identity and orientation** — Each inspection verifies that the correct component and orientation match the approved drawing before continuing.
- 8. B. Refer it to the teacher before continuing** — Uncertainty must be referred to the teacher so manufacturing continues from verified information rather than assumptions.
- 9. C. Using clear labels, photographs and annotated observations** — Traceability is strengthened by evidence linking the drawing, production steps and completed component.
- 10. A. The drawing, production plan, physical component and folio record** — Traceability links approved planning, the manufactured component and recorded evidence into one verified production history.

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Module 2 • answers 11–20

Student pages 24–25 / Shovel-head production

11. B. Ductility — Ductility describes shape change without failure, helping explain why some materials form more successfully than others.

12. A. Indentation, cutting or further deformation — Hardness influences resistance to indentation, cutting and further deformation rather than describing all forming behaviour.

13. B. Thickness — Thickness can influence resistance during forming, so observations should be linked to verified material information.

14. C. Material orientation — Where relevant to the confirmed material, orientation may influence forming consistency and should not be ignored.

15. A. A material becoming more resistant to further shaping after repeated deformation — Work hardening occurs after repeated deformation, increasing resistance to further shaping rather than reducing it.

16. B. A material partly returning towards its earlier form after shaping force is removed — Springback is partial recovery towards the earlier shape after the forming force is removed.

17. C. Record the response and refer it to the teacher — Unexpected behaviour should be recorded and referred to the teacher before making further processing decisions.

18. A. Changes in shape, resistance and surface condition — Observing shape, resistance and surface condition provides evidence about how material behaviour influenced the result.

19. B. Because an unexpected response does not prove a cause by itself — Evidence must support any explanation because an unexpected response alone does not establish its cause.

20. C. Teacher-issued material information, workshop drawing, SOP, WMS and demonstration — Authoritative teacher-issued information controls unresolved details instead of assumptions or previous workshop experience.

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Module 2 · answers 21–30

Student pages 26–27 / Shovel-head production

21. B. The teacher-issued drawing and confirmed quality criteria — Inspection should be based on confirmed drawings and criteria rather than judging appearance or comparing with other work.

22. A. Unsuitable edge condition — Edge condition can affect later fitting, so it should be checked against confirmed requirements before assembly proceeds.

23. B. Because these checks can influence one another — Inspection considers related features because a change in one area may affect alignment or later fitting.

24. A. The observed fault — Describe the observed fault first so later explanations remain evidence-based rather than assumptions.

25. B. Approved measurements, photographs and annotated notes — Reliable inspection evidence combines approved measurements, photographs and annotated notes linked to the inspection stage.

26. C. Document the doubt and refer it to the teacher — Uncertain results must be documented and reviewed by the teacher before later assembly stages continue.

27. B. Describe the distortion before suggesting handling as a possible cause — Observed evidence comes first, while possible causes should remain tentative unless supported by evidence.

28. A. The component, inspection stage and comparison reference — Inspection records should clearly identify the component, inspection stage and drawing or criterion used for comparison.

29. C. Readiness must be confirmed through inspection and teacher review — Progress depends on confirmed inspection results and teacher review, not the readiness of other components.

30. C. The teacher-issued drawing, quality criteria, WMS and demonstration — When details remain unresolved, the authorised drawing, quality criteria, WMS and demonstration provide the controlling evidence.

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Module 3 • answers 1–10

Student pages 32–33 / Lathe, fits and threads

- 1. A. Understanding of the authorised process** — Good SOP evidence explains understanding by linking the authorised process to the demonstrated machining task and drawing.
- 2. B. The SOP used, component and recognised control points** — Strong evidence identifies the relevant SOP, component or feature, and recognised control points from the teacher demonstration.
- 3. C. The current SOP and teacher demonstration** — Exact actions must come from the current authorised SOP and teacher demonstration, not memory or generic procedures.
- 4. A. Report the change and review the process** — Changed conditions require reporting and review because previous approval cannot automatically apply to a different situation.
- 5. A. The workpiece, setup or instruction changes** — Any change affecting the demonstrated conditions requires review before work continues under the authorised process.
- 6. B. Inspection against the verified drawing and teacher-controlled criteria** — Inspection confirms whether the machined result matches the verified drawing and approved acceptance criteria before later work.
- 7. C. Show observations, applied control points and whether review was needed** — Annotated photographs strengthen evidence by showing observations, recognised control points and any need for teacher review.
- 8. A. To demonstrate understanding of the evidence** — Connecting purpose, process and drawing demonstrates genuine understanding instead of repeating information without context.
- 9. B. A comparison with the verified drawing and approved criteria** — Inspection evidence should compare the machined result with verified drawing information and approved acceptance criteria.
- 10. C. Report the changed condition before proceeding** — Reporting changed conditions ensures the authorised process is reviewed before machining continues under different circumstances.

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Module 3 • answers 11–20

Student pages 34–35 / Lathe, fits and threads

- 11. B. The dimensional relationship between components that assemble or operate together** — A functional fit explains how related components dimensionally interact during assembly or operation, not how they look or were made.
- 12. A. Interference fit** — An interference fit requires tight engagement, unlike clearance or transition fits, which may allow some space or slight variation.
- 13. A. To understand how dimensions influence movement, alignment and assembly** — Dimensions should be interpreted through their functional role so machining supports correct assembly, movement and long-term performance.
- 14. B. The teacher-issued drawing and acceptance criteria** — General fit categories describe possibilities, but the required project fit comes from verified drawings and acceptance criteria.
- 15. A. To confirm the component remains suitable before more material is removed or assembly begins** — Progressive inspection helps identify developing issues early, reducing the risk of removing unnecessary material before assembly.
- 16. B. Because functional relationships must be checked against verified drawings and teacher-controlled criteria** — A component may appear correct yet still fail functional requirements, so verified drawing information and criteria must be used.
- 17. C. It may produce either slight clearance or slight interference** — A transition fit can result in either slight clearance or slight interference depending on the verified dimensional relationship.
- 18. A. Wear, maintenance and whether parts can later be separated or serviced** — Fit selection can affect wear, maintenance and future servicing as well as assembly and movement.
- 19. B. Document the issue and refer it to the teacher before attempting correction** — Unexpected fit problems should be documented and referred before correction so decisions remain based on verified project requirements.
- 20. C. Movement, alignment and serviceability** — Dimensional checks should always be related to how components move, align and can be serviced rather than appearance alone.

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Module 3 • answers 21–30

Student pages 36–37 / Lathe, fits and threads

21. A. The distance between corresponding points on neighbouring thread forms — Pitch measures spacing between neighbouring thread forms, helping students interpret the teacher-issued drawing rather than estimate by appearance.

22. B. It is formed on the outside of a component — An external thread is formed on the outside of a component. Drawings and teacher-issued information identify the required specification.

23. A. Appearance alone cannot confirm the specified form, size or condition — Successful engagement depends on the specified form, size and condition, which must be checked against approved sources and evidence.

24. C. The crest — The crest is the outermost part of the thread. Correct vocabulary helps students accurately interpret teacher-issued drawing information.

25. A. To compare the thread with the drawing and acceptance criteria — Approved checking methods provide evidence by comparing the thread with the drawing and acceptance criteria instead of relying on judgement.

26. B. Document the issue and refer it to the teacher before assembly — Resistance may indicate a mismatch or damage. Record the evidence and refer to the teacher before attempting assembly or correction.

27. B. The largest thread diameter — The major diameter is the largest thread diameter. Specific values come from the teacher-issued drawing and approved sources.

28. B. Apparently similar threads may not match the required specification — Matching form, size and condition control engagement. Always confirm the specified thread using teacher-issued drawings and approved checks.

29. A. Component identity, specified thread information, inspection evidence and any uncertainty — Recording inspection evidence and uncertainties creates a traceable record linked to the drawing and teacher-approved inspection process.

30. B. The thread is formed within an opening — An internal thread is formed within an opening. Use correct terminology to interpret the drawing before checking or assembly.

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Module 4 • answers 1–10

Student pages 42–43 / Pivot, milling and RPM

- 1. A. The relative movement path of related components** — The pivot centre determines how related components move together, so its confirmed position must match the teacher-issued drawing.
- 2. B. The teacher-issued drawings** — The teacher-issued drawings are the authoritative source for pivot details. Do not estimate centres from appearance or examples.
- 3. C. They help preserve the intended relationship between matching features** — Consistent datums and orientation marks help matching features remain correctly related throughout manufacture and assembly.
- 4. A. The differences may accumulate and affect movement during assembly** — Small alignment differences can build up across components and influence movement, so matching features should be checked carefully.
- 5. B. Because their condition and alignment may influence how parts sit and move together** — Contact face condition and alignment affect how components fit and move, making inspection an important evidence-based check.
- 6. C. From the approved drawing and teacher criteria** — Clearance must match the approved drawing and teacher criteria rather than being guessed or adjusted without review.
- 7. A. Record the issue and refer it to the teacher** — Uncertain clearance should be recorded and referred to the teacher because approved sources control the correct relationship.
- 8. B. To check identity, orientation, contact, alignment and movement while changes are still reversible** — Dry assembly provides a reversible opportunity to diagnose alignment and movement before permanent steps are completed.
- 9. C. Binding, unexpected movement or a mismatch** — Binding, unexpected movement or mismatched parts are diagnostic evidence that should be recorded and reviewed before proceeding.
- 10. A. The teacher-issued drawings, acceptance criteria, assembly plan and demonstration** — The authorised drawings, acceptance criteria, assembly plan and teacher demonstration control unresolved movement and pivot details.

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Module 4 • answers 11–20

Student pages 44–45 / Pivot, milling and RPM

11. A. The relationship between the rotating cutter and the secured workpiece — Milling depends on controlling the relationship between cutter and workpiece, not simply removing material or improving appearance.

12. B. To provide a consistent basis for locating and checking the work — A consistent datum links the milling operation to the teacher-issued drawing and supports accurate feature location and checking.

13. A. The result may no longer match the intended position or form — Movement changes the controlled relationship, so the developing feature may no longer match the intended drawing position or form.

14. C. Stop and follow the teacher-approved setup and demonstration — The approved setup and teacher demonstration control unresolved details. Do not substitute another setup or rely on assumptions.

15. A. To compare the developing feature with verified drawing information before removing more material — Progressive inspection checks developing work against verified drawing information before further material is removed.

16. B. The teacher-issued drawing, milling SOP, approved setup and demonstration — The teacher-issued sources are authoritative whenever operating, reference or inspection details are not explicitly stated.

17. A. Every approved setup relates to the confirmed workpiece, drawing and demonstrated process — Approved setups are linked to the confirmed drawing and workpiece, so another job's arrangement cannot be assumed suitable.

18. C. Report the changed condition immediately — Unexpected changes must be reported immediately because authorisation applies only to the confirmed setup and reference conditions.

19. A. Material removal depends on controlling the authorised relationship between tool and work — The theory emphasises controlled movement and positioning, not mere cutter contact, as the basis for accurate milling.

20. B. The teacher-issued drawing, milling SOP, approved setup and demonstration — Teacher-issued sources control movement paths, references and other unresolved details, so students should rely on those authoritative guides.

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Module 4 • answers 21–30

Student pages 46–47 / Pivot, milling and RPM

21. A. The teacher-provided data table for the nominated operation — Use only verified teacher-supplied data that applies to the nominated operation before completing the calculation.

22. B. To ensure the calculation uses consistent, teacher-approved units — Matching units helps ensure the approved formula is applied consistently using verified teacher-issued information.

23. C. Showing the formula, units, substituted values and working — Complete working provides evidence that verified values and the approved calculation process were followed correctly.

24. A. A theoretical spindle speed that still requires comparison with authorised settings — A calculated RPM is theoretical and must still be checked against authorised machine settings and teacher guidance.

25. B. Compare the result with available settings using the SOP and teacher demonstration — The current SOP and teacher demonstration determine the authorised machine setting when an exact match is unavailable.

26. C. Resolve the missing information before calculating or drilling — Missing information must be confirmed from teacher-issued sources before calculation or drilling continues.

27. A. They may not apply to the nominated drilling operation — Only teacher-approved values that apply to the nominated operation provide valid evidence for the calculation.

28. B. Only as directed by the teacher — Follow the teacher's directions for rounding so the calculation remains consistent with authorised expectations.

29. C. Verified values with units, the formula, substitutions and working — Recording each calculation stage shows the result came from verified data rather than guesswork or memory.

30. A. Refer to the teacher-provided formula, data table, SOP and demonstration before proceeding — Teacher-issued sources control unresolved calculations and authorised settings, so uncertainties must be resolved before continuing.

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Module 5 • answers 1–10

Student pages 52–53 / Assembly and evaluation

- 1. A. It checks whether completed components work together as required before final completion.** — The assembly confirms completed parts function together before finishing, using teacher-issued drawings and acceptance criteria as evidence.
- 2. B. Each part's identity, orientation and intended position.** — Checking identity, orientation and intended position ensures the assembly matches the authorised drawing before further decisions are made.
- 3. C. By using verified evidence rather than force or appearance.** — Fit should be supported by verified evidence, not assumptions based on appearance or forcing components together.
- 4. A. That the assembly follows its controlled movement path without unexpected resistance, contact or misalignment.** — Movement should match the authorised relationship shown in teacher-issued sources without unexpected resistance, contact or misalignment.
- 5. B. Inspect its condition and engagement using the approved method.** — Where specified, thread or fastener engagement must be checked using the approved method described by teacher-controlled sources.
- 6. C. To keep the work reversible while relationships between parts are checked.** — Trial assembly allows inspection and adjustment within the demonstrated process before irreversible finishing or handover occurs.
- 7. A. Rework changes a manufactured feature and requires teacher review and authorisation.** — Changing a manufactured feature is rework and cannot proceed without teacher review and authorisation.
- 8. B. Photographs, annotated observations and recorded checks identifying any uncertainty.** — Inspection evidence should clearly show what was checked and record any uncertainty requiring further review.
- 9. C. Only after teacher approval using the confirmed acceptance criteria.** — Teacher approval and confirmed acceptance criteria determine whether the assembly can proceed to finishing or handover.
- 10. A. The teacher-issued assembly drawing, acceptance criteria, SOP, WMS and demonstration.** — Where details remain unresolved, teacher-issued documents and demonstrations are the authoritative source for assembly decisions.

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Module 5 • answers 11–20

Student pages 54–55 / Assembly and evaluation

11. A. Evidence from the completed project and credible referenced sources — Impact judgements should come from verified project evidence and acknowledged credible sources, not assumptions or generalisations.

12. B. The categories are connected but each requires its own supporting evidence — Societal and environmental impacts may relate, but each conclusion still needs appropriate supporting evidence.

13. B. Explain benefits, limitations and supporting evidence — Balanced reports explain strengths, limitations and supporting evidence rather than making unsupported sustainability claims.

14. A. Actual project material records — Recorded project evidence is stronger than opinions or assumptions when judging resource use.

15. B. Because improvements in one area may increase impacts in another — Production decisions can improve function or serviceability while increasing material use, processing or waste.

16. B. State the limitation instead of inventing a conclusion — Evidence-first reporting recognises limitations instead of filling gaps with unsupported conclusions.

17. A. Production photographs, waste observations and documented quality checks — Recorded project evidence provides stronger support than opinions or observations without documentation.

18. A. Teacher-issued requirements, project records and approved referenced sources — Teacher-issued sources, verified project records and approved references control unresolved impact claims.

19. A. Evidence about construction quality and intended use — Claims about reduced replacement require supporting evidence linking durability with construction quality and intended use.

20. B. Because credible external facts must be properly referenced — Credible external information strengthens the report only when it is properly acknowledged alongside project evidence.

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Module 5 • answers 21–30

Student pages 56–57 / Assembly and evaluation

- 21. A. To consolidate the complete evidence chain for the project** — The report brings together verified planning, production and evaluation evidence rather than adding unrelated new material.
- 22. B. Planning, production, costing, quality and impact evidence** — The report should connect planning, production, costing, quality checks, impacts and evaluation into one supported evidence chain.
- 23. A. Explains the production stage, what the image proves and its connection to a decision or criterion** — Strong captions explain the evidence shown and link it directly to project decisions or assessment criteria.
- 24. C. Recorded to reflect what was actually used and made** — Authentic records should accurately reflect the completed project rather than estimates or planned values.
- 25. C. The folding action was smooth, improving usability, likely due to accurate alignment shown in recorded checks** — Each PMI point should describe an effect and, where supported by evidence, identify a likely cause.
- 26. B. Record them honestly as part of the as-built project** — Authentic submission requires honest reporting of faults, missing evidence and unresolved issues rather than concealing them.
- 27. B. Changes, assistance and unresolved approval should be acknowledged** — Authentic submissions accurately acknowledge assistance, changes and unresolved approval instead of presenting an idealised project.
- 28. A. The current Task 2 handbook, teacher-issued documents and teacher directions** — Teacher-issued sources control unresolved assessment details and should always be checked before making assumptions.
- 29. B. To show how evidence connects across the completed project** — Connecting related evidence demonstrates how planning, production and evaluation support the completed project as a whole.
- 30. C. Refer to teacher-issued sources because those details remain teacher to confirm until formally verified** — Where assessment details remain unresolved, students should rely on teacher-issued sources rather than assumptions or estimates.

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Module 1 • written evidence

Assess source traceability, understanding and authentic evidence; accept valid alternatives.

Evidence task • 10.1 • student page 9

Explain how two confirmed views/details combine to describe one component relationship.

Accept two identified views or details of the same component, what each view establishes and how the information combines. A strong handle example connects the assembly position with the pipe section and end-plug detail, without supplying an unsupported fit allowance. Require actual sheet titles and any uncertainty.

Evidence task • 10.2 • student page 10

Justify one WMS hold point that must be cleared before a later shovel process begins.

Accept a confirmed stage and later dependent process, a specific criterion and evidence, and why uncertainty must be resolved first. The record must name who clears the hold point and distinguish proposed planning from actual teacher approval. Example: an identified component remains on hold until its required feature has been checked against the current drawing; proceeding could hide an error inside the assembly. The component and check remain source-dependent.

Evidence task • 10.3 • student page 11

Frame one societal and one environmental research question tied to a confirmed shovel design/production decision.

Accept one question about people/use and one about resources/impacts, both tied to a confirmed decision. Require a named evidence source for each and avoid claims that the shovel is automatically sustainable, durable or repairable. A research question is not a completed finding.

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Module 2 · written evidence

Assess source traceability, understanding and authentic evidence; accept valid alternatives.

Evidence task · 11.1 · student page 19

Convert one confirmed drawing feature into a plan entry containing reference, operation category, check and evidence.

The entry should name a real source feature and reference, an operation category confirmed by the teacher, a relevant inspection hold point and evidence to retain. Accept alternatives consistent with the approved plan. Reject inferred allowances, tools or order presented as source facts.

Evidence task · 11.2 · student page 20

Explain one observed material response and the evidence needed before deciding on a teacher-approved adjustment.

Require a real observed response or clearly labelled demonstration observation, evidence about material/process/reference and a distinction between observation and proposed cause. No unverified heating, over-bending or other correction should be treated as authorised. For a planned observation, the student must say evidence is pending.

Evidence task · 11.3 · student page 21

Complete a quality record separating criterion, inspection evidence, result, possible cause and authorised next step.

Look for five clearly separated elements: sourced criterion; actual observation/measurement; comparison result; tentative cause supported by evidence or left unresolved; teacher-authorised next action. A student opinion alone cannot pass the part. Record pending approval honestly.

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Module 3 • written evidence

Assess source traceability, understanding and authentic evidence; accept valid alternatives.

Evidence task • 12.1 • student page 29

Explain the purpose of two teacher-confirmed SOP controls and the evidence that they were applied.

Accept two controls actually confirmed in the current SOP/demonstration, each with a clear purpose and suitable evidence. Evidence must not be fabricated or equated with general workshop permission. The student should identify the SOP, component and relevant teacher checkpoint.

Evidence task • 12.2 • student page 30

Explain how drawing values and measurements would support a teacher-confirmed fit judgement for one nominated pair.

Require the nominated component pair, current drawing requirement, appropriate measurements from both mating parts, comparison with teacher criteria and resulting evidence. The handle plug is explicitly called an interference fit, but no numeric allowance should be invented. Unexpected movement/assembly difficulty must be referred for review.

Evidence task • 12.3 • student page 31

Produce an inspection explanation showing how a specified thread would be identified, checked and recorded using teacher-approved evidence.

Look for component identity, written thread specification, current drawing reference, teacher-approved check method or gauge, recorded finding and review of any resistance/damage/uncertainty. Students should distinguish a symptom from a supported cause and avoid substituting an apparently similar fastener or forcing engagement. Actual pass/fail criteria remain teacher-controlled.

TEACHER GUIDE

Module 4 • written evidence

Assess source traceability, understanding and authentic evidence; accept valid alternatives.

Evidence task • 13.1 • student page 39

Explain a dry-assembly check that traces one observed movement issue back to measured or visual evidence.

Describe a teacher-approved dry-assembly check, identify one observed movement issue and relate it to a verified centre/contact/orientation or measurement reference. The cause remains tentative until supported. Do not reward invented operating positions, clearance values or adjustment methods.

Evidence task • 13.2 • student page 40

Describe how a drawing reference, confirmed setup and inspection hold point form a traceable milling record.

Expect a named drawing/reference feature, evidence of the teacher-confirmed setup and a relevant progressive hold point. Explain how the inspection checks the developing feature against the source. No invented speeds, feeds, depths or workholding should appear as authorised facts.

Evidence task • 13.3 • student page 48

Present a fully labelled calculation using teacher-supplied values, then state how the authorised available setting is confirmed and recorded.

Verify against the actual teacher-supplied data and formula. Require source, values with units, correct substitution, transparent working, directed rounding and an explicit distinction between theoretical RPM and the confirmed available setting. Accept a pending-data record when the required values/approval genuinely have not been supplied; do not fabricate a number or sign-off.

TEACHER GUIDE

Module 5 • written evidence

Assess source traceability, understanding and authentic evidence; accept valid alternatives.

Evidence task • 14.1 • student page 49

Write an assembly quality-gate record covering confirmed criterion, movement evidence, issue diagnosis and teacher-confirmed action.

Look for confirmed assembly criterion, component/orientation identification, actual movement evidence, a diagnosis supported by observations or measurements, and teacher-confirmed action or unresolved review. Distinguish reversible trial assembly from rework. No student-only declaration of safety or acceptance is sufficient.

Evidence task • 14.2 • student page 50

Write one supported impact paragraph containing claim, evidence/source, limitation and realistic improvement.

Require one bounded societal/environmental claim, specific project evidence or credible cited source, explanation of the connection, a limitation and a realistic improvement. Accept justified alternatives. Reject universal sustainability claims, invented observations or unsupported lifetime estimates.

Evidence task • 14.3 • student page 51

Draft a concise final report conclusion that integrates quality, impact, cost/production evidence, limitation and next-step improvement.

A concise conclusion should integrate quality, impacts and actual production/cost evidence, state a limitation and identify a realistic improvement. It should acknowledge assistance, changes and missing/pending evidence. Assessment dates/marks/submission requirements must be confirmed from the current teacher notice, not inferred from this workbook.

TEACHER GUIDE

Applied learning · response guidance

The examples below are criteria and model reasoning, not invented student results.

Student page 18 · p14-transfer

Accept a bounded research question linked to one decision and consequence. For the recycling claim, a useful question asks how the actual material could be recovered locally after its useful life; evidence could include the confirmed material and the recovery service requirements. A missing material identification or unavailable local route is a limitation. For durability, compare documented maintenance/quality evidence and expected use; do not accept an unsupported service-life claim.

Student page 41 · p29-interpret

The factor 1000 converts cutting speed from metres per minute to millimetres per minute, matching the drill diameter units. The result is theoretical, using illustrative data; actual material/operation data, available settings, current SOP and teacher authorisation must be confirmed separately.

Student page 28 · p20-audit

B is the strongest starting point: it names an observation and confirmed reference and keeps the cause unresolved. It still needs the exact source criterion and supporting inspection detail before a decision. A offers appearance alone, with no criterion or measurement and an unsupported readiness judgement. C proposes a cause without evidence and an unauthorised correction. Credit explanations that clearly separate observation, criterion, cause and authorisation.

TEACHER GUIDE

Applied learning · response guidance

The examples below are criteria and model reasoning, not invented student results.

Student page 28 · p20-repair

Accept a record that identifies the component, records only the actual hypothetical observation, names the source/check still required and holds the decision for teacher review. Example: 'The surface appears neat. Geometry and feature position have not yet been checked against the current head drawing; assembly readiness remains pending the approved inspection.' Do not accept fabricated dimensions or approval.

Student page 38 · p26-tree

Valid branches may include wrong specification (check current drawing and component identity), damaged/contaminated thread condition (approved inspection), incomplete geometry (specified gauge/check), or incorrect orientation/related fit (assembly reference and approved comparison). A branch is a hypothesis, not a diagnosis. Methods remain teacher-confirmed. Do not accept forcing engagement as a test or correction.

Student page 38 · p26-judgement

A functional fit depends on the specified relationship, form/size and condition of both mating parts, and evidence against the acceptance criteria. A visible resemblance cannot establish all of those. The source, check result, proposed action and practical permission must be reviewed by the teacher before alteration or continuation.

TEACHER GUIDE

Applied learning · response guidance

The examples below are criteria and model reasoning, not invented student results.

Student page 48 · p32-rpm-practice

$n = (1000 \times 15) \div (\pi \times 12) = 15000 \div 37.699... = 397.887... \text{ RPM}$, or 397.9 RPM to one decimal place. Units are consistent with v_c in m/min and D in mm. The result is theoretical and uses illustrative inputs; actual data, machine options, SOP and teacher authorisation are separate requirements.

Student page 58 · p38-chain

Require a specific criterion/question, genuine evidence, a reasoned interpretation, an honest limitation and an achievable improvement. Each link must follow from the previous one; appearance or preference alone cannot support a functional or environmental claim. Missing evidence should remain visibly pending.

Student page 58 · p38-authenticity

Credit a candid audit that names unsupported parts of the chain, accurately attributes sources/demonstrations/assistance, and identifies the actual current submission or approval checkpoint. A clean-looking report with hidden omissions is not a stronger answer.

TEACHER GUIDE

Folio 1 • Project brief and function

Student page 59 / Judge the actual source, decision and evidence.

Summarise the need, intended user, function and teacher-issued requirements.

Core evidence

A valid brief uses the actual teacher-issued need, intended user and function, then connects requirements with evidence. Credit clearly labelled uncertainties. Reject invented capacities, loads, operating positions or claims copied from an unrelated commercial shovel.

Evidence caption

The caption identifies the actual evidence, explains what it shows and links it to a criterion or decision. It should not claim an outcome that has not been observed, or treat a photograph as proof of an entire process.

Source / teacher checkpoint

Name the actual drawing, brief, SOP, approved data source, demonstration or feedback. Record the relevant title/date or checkpoint and distinguish a completed confirmation from one still pending.

The readiness checkbox is a student record, not a practical sign-off. An optional authorised photograph can support evidence; it cannot replace the required explanation and source record.

TEACHER GUIDE

Folio 2 · Requirements and modifications

Student page 60 / Judge the actual source, decision and evidence.

Separate requirements from constraints and explain any teacher-approved modification.

Core evidence

Require accurate separation of requirement, constraint and preference, with a source and practical implication. A proposed change must not be reported as approved. If no modification is relevant, a source-based explanation of what should remain unchanged is valid.

Explanation / extension

Require accurate separation of requirement, constraint and preference, with a source and practical implication. A proposed change must not be reported as approved. If no modification is relevant, a source-based explanation of what should remain unchanged is valid.

Evidence caption

The caption identifies the actual evidence, explains what it shows and links it to a criterion or decision. It should not claim an outcome that has not been observed, or treat a photograph as proof of an entire process.

Source / teacher checkpoint

Name the actual drawing, brief, SOP, approved data source, demonstration or feedback. Record the relevant title/date or checkpoint and distinguish a completed confirmation from one still pending.

The readiness checkbox is a student record, not a practical sign-off. An optional authorised photograph can support evidence; it cannot replace the required explanation and source record.

TEACHER GUIDE

Folio 3 • WHS and risk controls

Student page 61 / Judge the actual source, decision and evidence.

Record hazards, risk level, stronger controls, PPE and teacher permission points.

Core evidence

Look for a task-specific hazard, current teacher-approved risk/control information, stronger controls where applicable, PPE within its proper role and a clear permission/review point. The record must not invent operating steps or present a completed worksheet as workshop approval.

Explanation / extension

Look for a task-specific hazard, current teacher-approved risk/control information, stronger controls where applicable, PPE within its proper role and a clear permission/review point. The record must not invent operating steps or present a completed worksheet as workshop approval.

Evidence caption

The caption identifies the actual evidence, explains what it shows and links it to a criterion or decision. It should not claim an outcome that has not been observed, or treat a photograph as proof of an entire process.

Source / teacher checkpoint

Name the actual drawing, brief, SOP, approved data source, demonstration or feedback. Record the relevant title/date or checkpoint and distinguish a completed confirmation from one still pending.

The readiness checkbox is a student record, not a practical sign-off. An optional authorised photograph can support evidence; it cannot replace the required explanation and source record.

TEACHER GUIDE

Folio 4 • Materials investigation

Student page 62 / Judge the actual source, decision and evidence.

Compare relevant metals or stock forms and justify the teacher-specified selection.

Core evidence

Expect common criteria, credible material/form evidence and a reasoned justification tied to the confirmed project choice. Properties, stock shape and process behaviour should not be treated as interchangeable. Material grade, thickness and substitutions remain source-dependent.

Explanation / extension

Expect common criteria, credible material/form evidence and a reasoned justification tied to the confirmed project choice. Properties, stock shape and process behaviour should not be treated as interchangeable. Material grade, thickness and substitutions remain source-dependent.

Evidence caption

The caption identifies the actual evidence, explains what it shows and links it to a criterion or decision. It should not claim an outcome that has not been observed, or treat a photograph as proof of an entire process.

Source / teacher checkpoint

Name the actual drawing, brief, SOP, approved data source, demonstration or feedback. Record the relevant title/date or checkpoint and distinguish a completed confirmation from one still pending.

The readiness checkbox is a student record, not a practical sign-off. An optional authorised photograph can support evidence; it cannot replace the required explanation and source record.

TEACHER GUIDE

Folio 5 • Research and ideas

Student page 63 / Judge the actual source, decision and evidence.

Record research, freehand sketches or modification ideas and explain the approved direction.

Core evidence

A useful research record identifies a source, extracts a relevant idea, labels uncertainty and explains the selected direction or pending approval. Do not accept copied commercial geometry as a new specification. Original sketches may explore ideas without authorising them.

Explanation / extension

A useful research record identifies a source, extracts a relevant idea, labels uncertainty and explains the selected direction or pending approval. Do not accept copied commercial geometry as a new specification. Original sketches may explore ideas without authorising them.

Evidence caption

The caption identifies the actual evidence, explains what it shows and links it to a criterion or decision. It should not claim an outcome that has not been observed, or treat a photograph as proof of an entire process.

Source / teacher checkpoint

Name the actual drawing, brief, SOP, approved data source, demonstration or feedback. Record the relevant title/date or checkpoint and distinguish a completed confirmation from one still pending.

The readiness checkbox is a student record, not a practical sign-off. An optional authorised photograph can support evidence; it cannot replace the required explanation and source record.

TEACHER GUIDE

Folio 6 · Working drawing evidence

Student page 64 / Judge the actual source, decision and evidence.

Explain how the teacher-issued drawing communicates material, dimensions, shape, tolerance or assembly information.

Drawing evidence

A useful sketch connects the same named component across two relevant original views or sheets, preserves the relationship without guessing dimensions, labels the source and separates confirmed information from unresolved points. Drawing finish alone does not establish accuracy.

Core evidence

The explanation should say what each view contributes and how cross-referencing reduces ambiguity. Accept material, dimension, shape, tolerance or assembly information only where present on the actual cited sheet. An uncertainty must be recorded for clarification rather than estimated.

Evidence caption

Caption explains what the sketch or authorised photograph demonstrates and links it to the chosen component relationship. It must not claim a manufactured result or teacher approval that is not evidenced.

Source / teacher checkpoint

Record the actual drawing title/date and relevant view or callout; distinguish the original 2014 drawing from any later teacher-issued version. Identify the teacher clarification only when it has occurred.

The readiness checkbox is a student record, not a practical sign-off. An optional authorised photograph can support evidence; it cannot replace the required explanation and source record.

TEACHER GUIDE

Folio 7 • Cutting, costing and resources

Student page 65 / Judge the actual source, decision and evidence.

Summarise your spreadsheet list and explain one decision that reduced waste or improved efficiency.

Core evidence

Look for traceable spreadsheet entries, actual quantities/cost sources and a justified efficiency or waste decision. Accept a pending-data record where values are not available; do not accept fabricated stock allowances, prices or savings. Formula errors should be corrected transparently.

Explanation / extension

Look for traceable spreadsheet entries, actual quantities/cost sources and a justified efficiency or waste decision. Accept a pending-data record where values are not available; do not accept fabricated stock allowances, prices or savings. Formula errors should be corrected transparently.

Evidence caption

The caption identifies the actual evidence, explains what it shows and links it to a criterion or decision. It should not claim an outcome that has not been observed, or treat a photograph as proof of an entire process.

Source / teacher checkpoint

Name the actual drawing, brief, SOP, approved data source, demonstration or feedback. Record the relevant title/date or checkpoint and distinguish a completed confirmation from one still pending.

The readiness checkbox is a student record, not a practical sign-off. An optional authorised photograph can support evidence; it cannot replace the required explanation and source record.

TEACHER GUIDE

Folio 8 • Production plan and WMS

Student page 66 / Judge the actual source, decision and evidence.

Sequence key processes, evidence checkpoints, demonstrations and approvals.

Core evidence

Require sourced stages, dependency, confirmed control/hold point, evidence and responsible teacher checkpoint. Sequence must match the current approved plan. The explanation should show why unchecked work cannot pass into the next stage and how changes are reviewed.

Explanation / extension

Require sourced stages, dependency, confirmed control/hold point, evidence and responsible teacher checkpoint. Sequence must match the current approved plan. The explanation should show why unchecked work cannot pass into the next stage and how changes are reviewed.

Evidence caption

The caption identifies the actual evidence, explains what it shows and links it to a criterion or decision. It should not claim an outcome that has not been observed, or treat a photograph as proof of an entire process.

Source / teacher checkpoint

Name the actual drawing, brief, SOP, approved data source, demonstration or feedback. Record the relevant title/date or checkpoint and distinguish a completed confirmation from one still pending.

The readiness checkbox is a student record, not a practical sign-off. An optional authorised photograph can support evidence; it cannot replace the required explanation and source record.

TEACHER GUIDE

Folio 9 • Production evidence

Student page 67 / Judge the actual source, decision and evidence.

Write captions for your own photographs that explain the process, control and result shown.

Core evidence

Captions need a specific process stage, actual observed control/result and a link to a decision or criterion. They must distinguish genuine evidence from planned photographs and avoid claiming that a still image proves an entire safe process. Respect authorised photography and privacy.

Explanation / extension

Captions need a specific process stage, actual observed control/result and a link to a decision or criterion. They must distinguish genuine evidence from planned photographs and avoid claiming that a still image proves an entire safe process. Respect authorised photography and privacy.

Evidence caption

The caption identifies the actual evidence, explains what it shows and links it to a criterion or decision. It should not claim an outcome that has not been observed, or treat a photograph as proof of an entire process.

Source / teacher checkpoint

Name the actual drawing, brief, SOP, approved data source, demonstration or feedback. Record the relevant title/date or checkpoint and distinguish a completed confirmation from one still pending.

The readiness checkbox is a student record, not a practical sign-off. An optional authorised photograph can support evidence; it cannot replace the required explanation and source record.

TEACHER GUIDE

Folio 10 • Quality and tolerance checks

Student page 68 / Judge the actual source, decision and evidence.

Record the approved criteria, measurement or fit evidence, result and any correction.

Core evidence

Require the sourced criterion, approved checking method, actual evidence, comparison result and clear next action. Possible causes remain tentative until supported. Passing one feature does not establish whole-product acceptance; pending teacher approval should be visible.

Explanation / extension

Require the sourced criterion, approved checking method, actual evidence, comparison result and clear next action. Possible causes remain tentative until supported. Passing one feature does not establish whole-product acceptance; pending teacher approval should be visible.

Evidence caption

The caption identifies the actual evidence, explains what it shows and links it to a criterion or decision. It should not claim an outcome that has not been observed, or treat a photograph as proof of an entire process.

Source / teacher checkpoint

Name the actual drawing, brief, SOP, approved data source, demonstration or feedback. Record the relevant title/date or checkpoint and distinguish a completed confirmation from one still pending.

The readiness checkbox is a student record, not a practical sign-off. An optional authorised photograph can support evidence; it cannot replace the required explanation and source record.

TEACHER GUIDE

Folio 11 • Industry and impacts

Student page 69 / Judge the actual source, decision and evidence.

Compare one classroom and industrial process, then explain a societal or environmental impact.

Core evidence

A fair comparison uses a common criterion and credible source for the industrial claim. The impact paragraph should connect actual or referenced evidence to a bounded conclusion, state a limitation and offer an achievable improvement. Avoid blanket claims about automation, recycling or sustainability.

Explanation / extension

A fair comparison uses a common criterion and credible source for the industrial claim. The impact paragraph should connect actual or referenced evidence to a bounded conclusion, state a limitation and offer an achievable improvement. Avoid blanket claims about automation, recycling or sustainability.

Evidence caption

The caption identifies the actual evidence, explains what it shows and links it to a criterion or decision. It should not claim an outcome that has not been observed, or treat a photograph as proof of an entire process.

Source / teacher checkpoint

Name the actual drawing, brief, SOP, approved data source, demonstration or feedback. Record the relevant title/date or checkpoint and distinguish a completed confirmation from one still pending.

The readiness checkbox is a student record, not a practical sign-off. An optional authorised photograph can support evidence; it cannot replace the required explanation and source record.

TEACHER GUIDE

Folio 12 · PMI and final evaluation

Student page 70 / Judge the actual source, decision and evidence.

Use plus, minus and interesting observations to judge function, quality and process, then identify one realistic improvement.

Core evidence

Each PMI point should be specific and supported, with an effect and a likely cause only where evidence justifies it. The conclusion integrates the actual project, recognises limitations and proposes a realistic improvement. Acknowledge sources/assistance and do not conceal faults or pending approval.

Explanation / extension

Each PMI point should be specific and supported, with an effect and a likely cause only where evidence justifies it. The conclusion integrates the actual project, recognises limitations and proposes a realistic improvement. Acknowledge sources/assistance and do not conceal faults or pending approval.

Evidence caption

The caption identifies the actual evidence, explains what it shows and links it to a criterion or decision. It should not claim an outcome that has not been observed, or treat a photograph as proof of an entire process.

Source / teacher checkpoint

Name the actual drawing, brief, SOP, approved data source, demonstration or feedback. Record the relevant title/date or checkpoint and distinguish a completed confirmation from one still pending.

The readiness checkbox is a student record, not a practical sign-off. An optional authorised photograph can support evidence; it cannot replace the required explanation and source record.